

• Easy • #27 • Remove Element

▲ Problem Statement • Given an array "nums" & a value "val", remove all occurrences of "val" in-place & return the number of remaining elements.

▲ Main Solution • Use two pointers :-
• "i" checks every element
• "j" keeps track of where to place the next element that is not equal to "val".

▲ Brute Force • Create another array & copy only the elements that are not equal to "val".

▲ Algorithm • 1 | Set "j = 0"

2 | Loop through the array using "i".

3 | If "nums[i] != val" :- Copy "nums[i]" to "nums[j]"

• Increment "j"

4 | If "nums[i] == val", skip it.

5 | Return "j".

▲ Important Points -

- No extra array is needed

- "i" checks every element.

- "j" is the next position for a valid element.

- Skip elements equal to "val"

- At the end, "j" is the number of remaining elements.

▲ Complexity -

- Brute Force $\rightarrow O(n) \& O(n)$
- Main Solution $\rightarrow O(n) \& O(1)$